Web-based Restaurant Management Guidelines

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Requirements

Install the following software before setting up the project:

- Python 3.10 or newer
- Node.js 18 or newer (npm is included with Node.js)
- PostgreSQL 14 or newer
- Git (optional, for source-control workflows)

Make sure the PostgreSQL service is running before starting the backend.

Installation

1. Clone the repository

   Open IDE terminal and clone the repository to your decided location: https://github.com/Nam3007/web-project

```
PS C:\Users\bquoc\Downloads> git clone https://github.com/Nam3007/web-project.git
```

   Or download project as zip

2. Install BE Dependencies

- First you will need to install the virtual environment:
  - From your terminal, cd to BE folder then enter following command:

    python -m venv .venv
  - Then active the virtual environment:
    - for powershell: **.\venv\scripts\activate.ps1**
    - for command prompt: **.\venv\scripts\activate.bat**
  - After activating virtual environmnet, install required dependencies:

    pip install -r requirements.txt

3. Install FE Dependencies

   Create another terminal instance and cd to FE folder and use **npm install** to install required dependencies for FE

Setup environments

1. **Create PostgreSQL database**

   Create a database named ‘restaurant’ and ensure schema named ‘restaurant’ exist

2. **Configure BE environment variable**

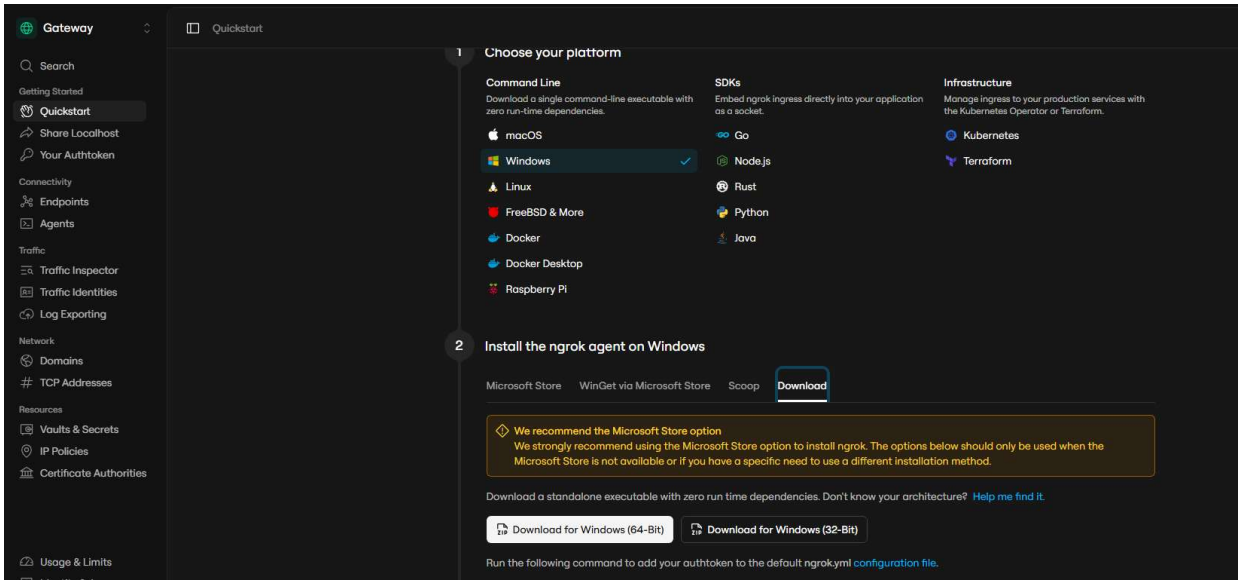
Create `BE\.env` locally. Do not commit this file or copy credentials into documentation. Use values appropriate for your PostgreSQL and payment-service accounts:

```dotenv
DATABASE_URL=postgresql://<user>:<password>@localhost:5432/restaurant
SECRET_KEY=<long-random-secret>
VIETQR_BANK_ID=<bank-id>
VIETQR_ACCOUNT_NO=<account-number>
VIETQR_ACCOUNT_NAME=<account-name>
SEPAY_WEBHOOK_API_KEY=<webhook-key>
```

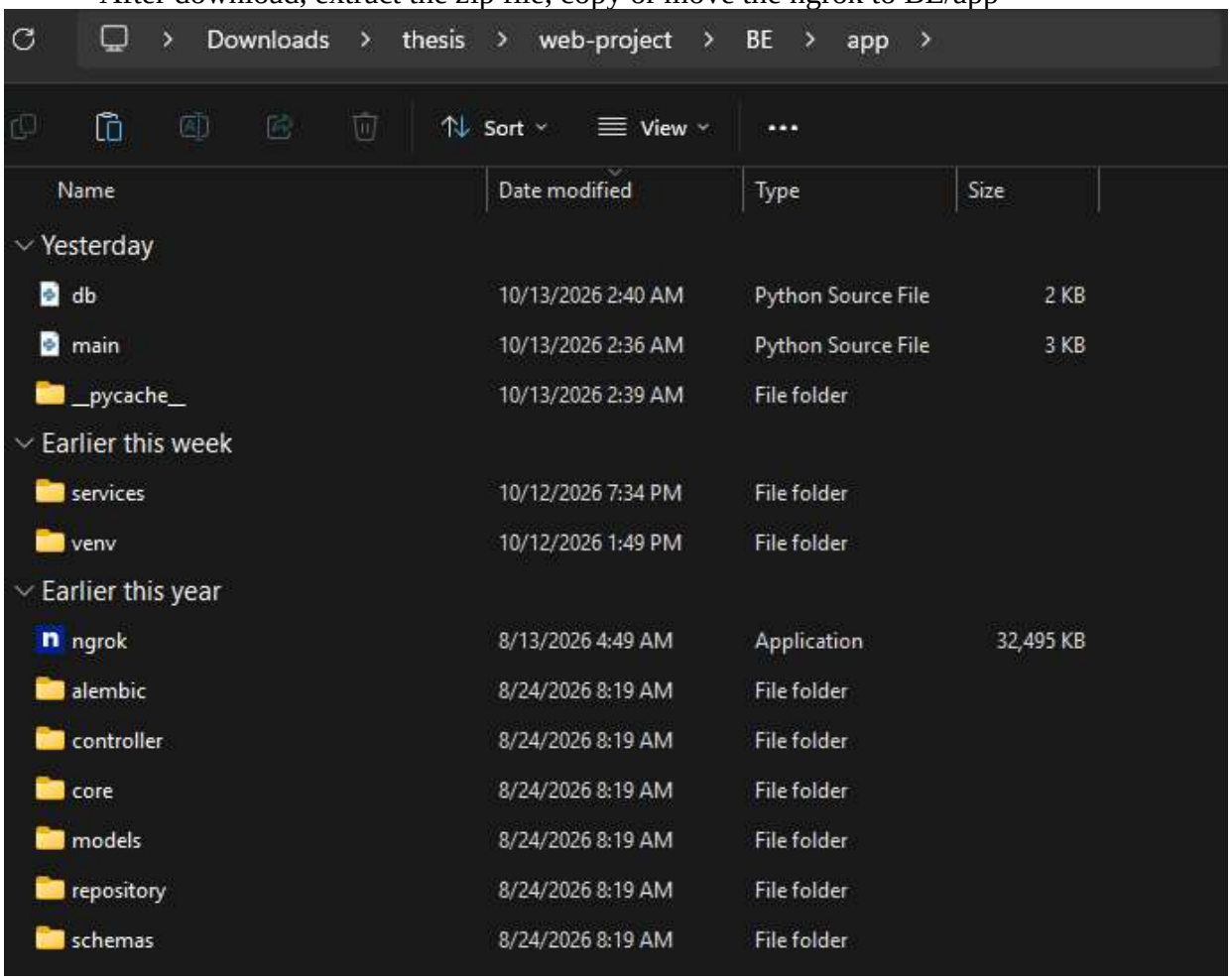
`DATABASE_URL` and `SECRET_KEY` are required. The VietQR and SePay values are needed when testing automated payment features; the application provides defaults for some VietQR settings.

3. **Sepay and ngrok setup (optional)**

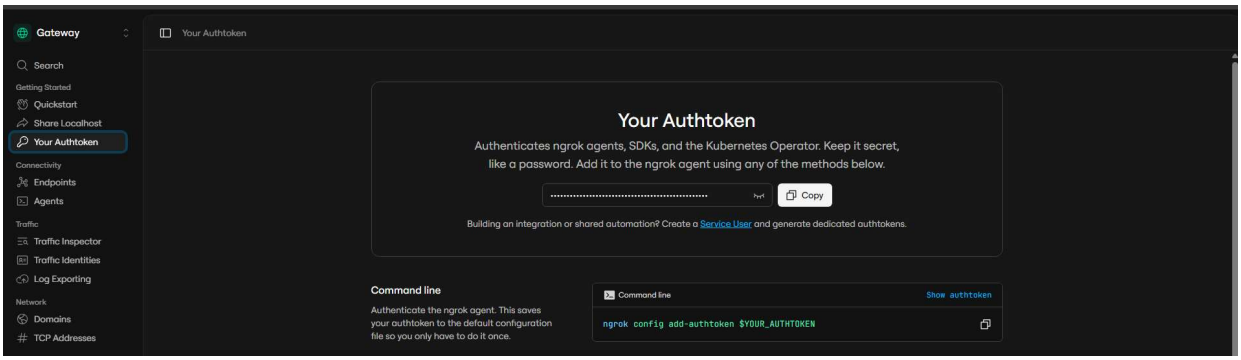
- ngrok (skip this part if deploy product)
  Since this project run locally so we will need external tool to forward the BE server to the internet so Sepay can connect to it.
  - Go to https://ngrok.com/ and create account
  - After create account, go to “Quick Start” tab and choose a download for your system.



- After download, extract the zip file, copy or move the ngrok to BE/app



- Go to “Your Authenticate” tab, copy the ngrok command line and create a new instance terminal to BE/app and paste to it




```
C:\Users\bquoc\Downloads\thesis\web-project\BE\app>ngrok config add-authtoken 3Kb
Authtoken saved to configuration file: C:\Users\bquoc\AppData\Local\ngrok\ngrok.yml
```

- Now use ngrok to forward the FastAPI port (run FastAPI server first before step) , the port of FastAPI might various but usually 8000

```
C:\Users\bquoc\Downloads\thesis\web-project\BE\app>ngrok http 8000
```

```
ngrok

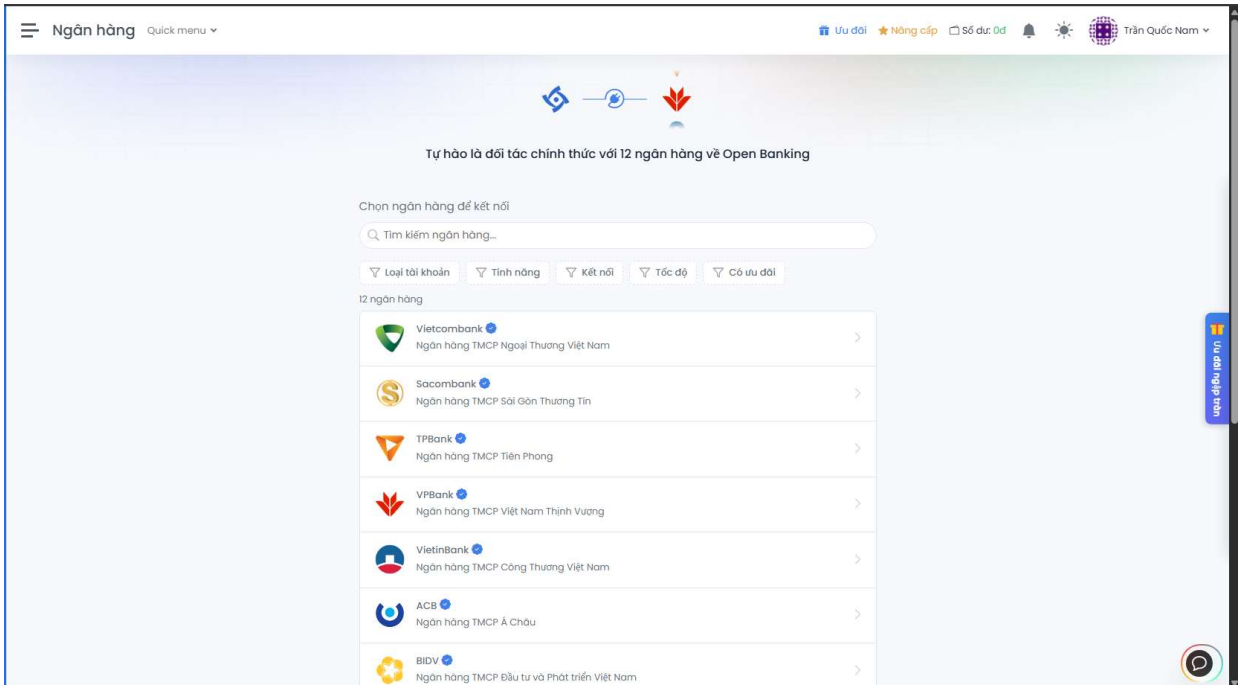
Jev is on ngrok.ai

Session Status                online
Account                       bquocnam3007@gmail.com (Plan: Free)
Version                       3.39.11
Region                        Asia Pacific (ap)
Web Interface                 http://127.0.0.1:4040
Forwarding                    https://handyman-unfazed-eclipse.ngrok-free.dev -> http://localhost:8000

Connections                   ttl     opn     rt1     rt5     p50     p90
                              0       0       0.00    0.00    0.00    0.00
```

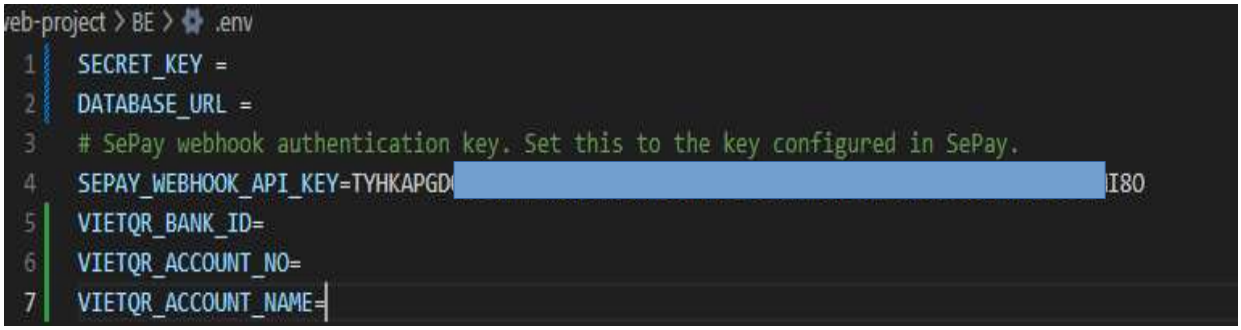
- copy the Forwarding URL, in this case is https://handyman-unfazed-eclipse.ngrok-free.dev

- Sepay:
  - Go to https://sepay.vn/ and create a new account; skip this part if you already have an account
  - Link your bank account with your sepay account

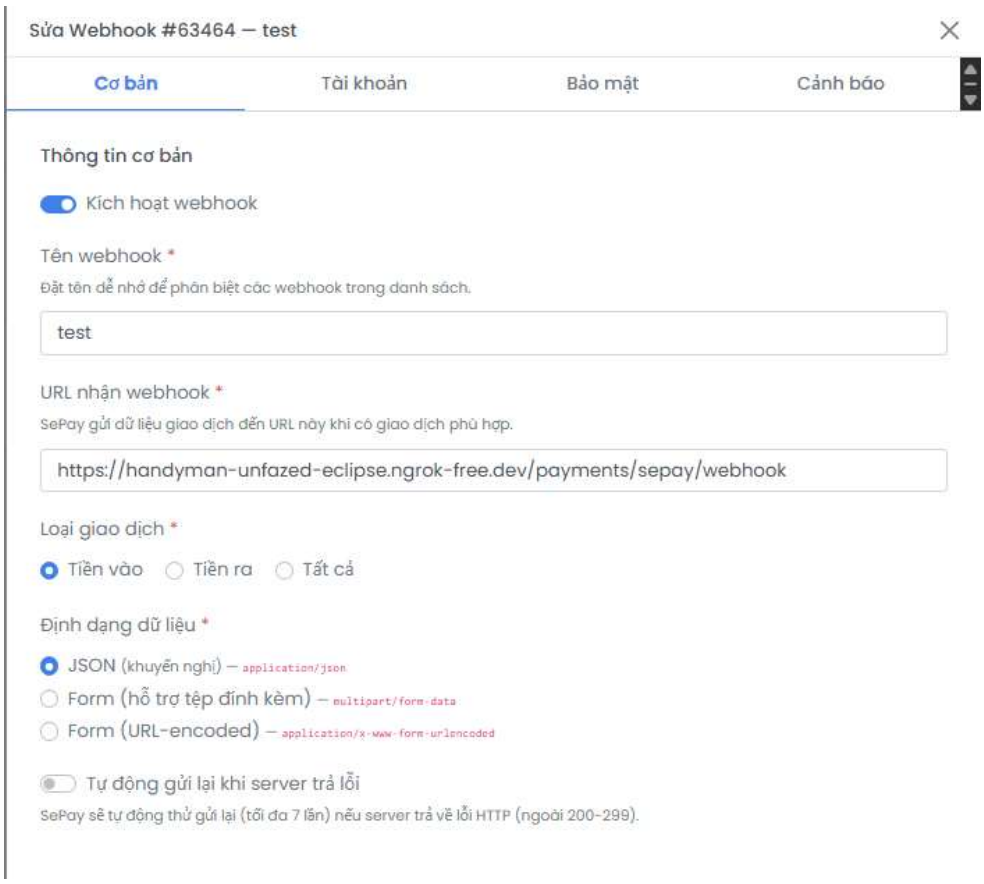


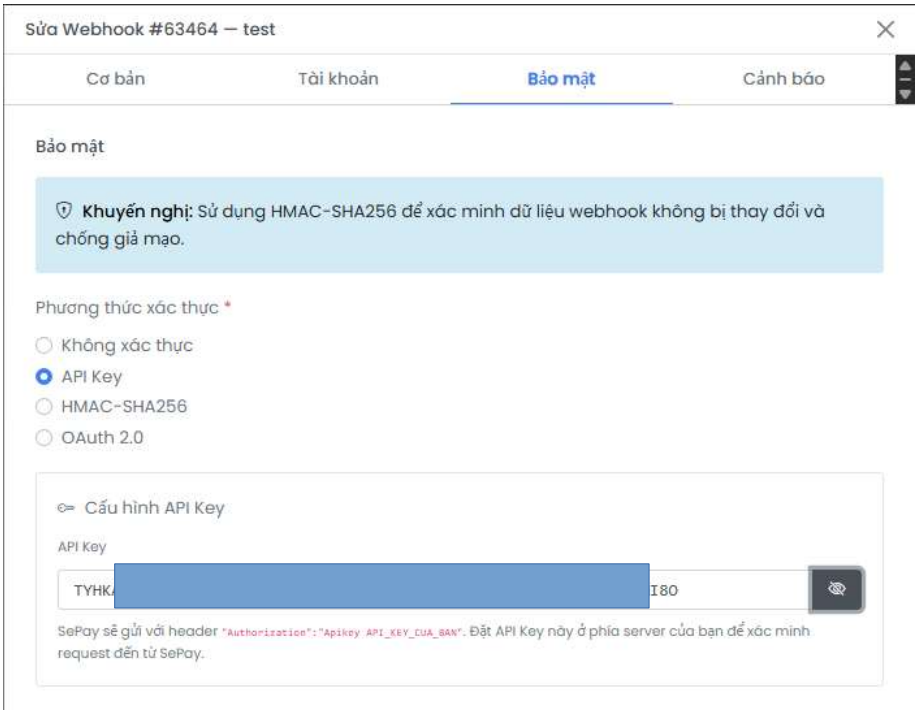
- Now create and API key and paste to .env and BE folder





- Now create a webhook and link to Fordwarding URL with /payment/sepay/webhook along with API key


Sửa Webhook #63464 — test
Cơ bản
Tài khoản
Bảo mật
Cảnh báo
Thông tin cơ bản
Kích hoạt webhook
Tên webhook *
Đặt tên dễ nhớ để phân biệt các webhook trong danh sách.
test
URL nhận webhook *
SePay gửi dữ liệu giao dịch đến URL này khi có giao dịch phù hợp.
https://handyman-unfazed-eclipse.ngrok-free.dev/payments/sepay/webhook
Loại giao dịch *
Tiền vào
Tiền ra
Tất cả
Định dạng dữ liệu *
JSON (khuyến nghị) — application/json
Form (hỗ trợ tệp đính kèm) — multipart/form-data
Form (URL-encoded) — application/x-www-form-urlencoded
Tự động gửi lại khi server trả lỗi
SePay sẽ tự động thử gửi lại (tối đa 7 lần) nếu server trả về lỗi HTTP (ngoài 200-299).


Sửa Webhook #63464 — test
Cơ bản
Tài khoản
Bảo mật
Cảnh báo
Bảo mật
Khuyến nghị: Sử dụng HMAC-SHA256 để xác minh dữ liệu webhook không bị thay đổi và chống giả mạo.
Phương thức xác thực *
Không xác thực
API Key
HMAC-SHA256
OAuth 2.0
Cấu hình API Key
API Key
TYHK
I80
SePay sẽ gửi với header "Authorization":"Apikey API_KEY_CUA_BAN". Đặt API Key này ở phía server của bạn để xác minh request đến từ SePay.

Run application

1. Start BE server

   From previous terminal that you have activated virtual environment, cd to /app and activate FastAPI server by using:
   FastAPI dev main.py

2. Start FE server

   You can either create another terminal instance and cd to FE or from previous terminal instance that you have used **npm install**, activate react vite server by using :
   npm run dev